

Shresth Rohan

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science with specialisation in Data Science
Current CGPA: **8.76** (till semester 4)

Vellore, Tamil Nadu
July 2024 – May 2028

St. Michael's High School

Class XII (CBSE)- 90.2%

Patna, Bihar
May 2024

Don Bosco Academy

Class X (ICSE)- 96.8%

Patna, Bihar
May 2022

PROJECTS

FocusGuard | Python, PyTorch, React, Node.js, MongoDB, ONNX

Feb. 2026– Present

- Built a full-stack AI attention monitoring system using MobileNetV2 transfer learning to classify user focus states (focused, distracted, drowsy) in real time via webcam
- Trained and exported a custom **ONNX model** achieving live inference through a **Flask REST API** with **MediaPipe** face detection and temporal smoothing
- Developed a React dashboard with session analytics, productivity scoring, and real-time confidence visualization backed by a **Node.js/Express API** and **MongoDB**
- Implemented a **Chrome extension** using Mutation Observer to **dynamically filter distracting DOM** elements across sites including YouTube, Instagram, and Twitter

Smart Adaptive Street Lighting System (Patent Filed) | ESP32, Arduino IDE, C++, IoT Sensors April 2025 – Present

- Developed an **ESP32-based intelligent street lighting system** integrating LDR, rain, sound, ultrasonic sensors, and a servo motor for adaptive illumination based on environmental and traffic conditions.
- Implemented **dynamic brightness control** using ambient light, rainfall detection, and emergency siren recognition to improve energy efficiency and prioritize emergency vehicle movement.
- Designed a **servo-assisted ultrasonic scanning mechanism** for vehicle detection, distance measurement, and overspeed monitoring with real-time buzzer alerts for enhanced road safety.
- **Filed an Indian patent application** for the proposed multi-sensor smart street lighting system, introducing an integrated approach to adaptive lighting, traffic monitoring, and smart city infrastructure.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS

Frameworks: React, Node.js, Express, Sprint Boot, Flask

AI/ML: PyTorch, ONNX, MediaPipe, pandas, NumPy, Matplotlib

Databases: MongoDB, PostgreSQL

Developer Tools: Git, Docker, VS Code

ACHIEVEMENTS & EXTRA CURRICULAR ACTIVITIES:

Secured **3rd** in State Rank and **239th** International Rank in *SilverZone Foundation International Informatics Olympiad*

Achieved **Zonal Rank 1 and International Rank 85** in *SOF International English Olympiad*

Community Service:

- Karo Sambhav Movement- *Cohesive E-Waste Movement Volunteer*
- VIT Outreach Program – *Taught Middle Schoolers about importance of Computers and Coding*

Interests:

- Game Development (*OpenGL, Unreal Engine 5*),
- UI/UX Design (*Figma, Canva*),
- Competitive Programming – Problem Solving